MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, CROSSLINKED, MODIFIED POLYETHYLENE (XLPE) INSULATED, 125°C, 1000 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.

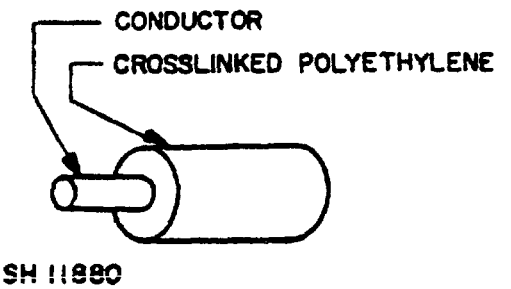


FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material	Coating		Min	Max
26	1 X 26	Copper	Tin	0.0159	0.046	0.054
26	7 X 34	Copper	Tin	.019	.049	.057
26	19 X 38	Copper	Tin	.020	.049	.057
24	1 X 24	Copper	Tin	.0201	.050	.058
24	7 X 32	Copper	Tin	.024	.054	.062
24	19 X 36	Copper	Tin	.025	.054	.062
22	1 X 22	Copper	Tin	.0254	.055	.064
22	7 X 30	Copper	Tin	.030	.060	.068
22	19 X 34	Copper	Tin	.032	.060	.068

FSC 6145

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TABLE I. Construction details.- Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
					Diameter (inch)	
		Material	Coating		Min	Max
20	1 X 20	Copper	Tin	0.0320	0.062	0.070
20	7 X 28	Copper	Tin	.038	.068	.076
20	19 X 32	Copper	Tin	.040	.068	.076
18	1 X 18	Copper	Tin	.0403	.070	.079
18	7 X 26	Copper	Tin	.048	.076	.086
18	19 X 30	Copper	Tin	.047	.076	.086
16	1 X 16	Copper	Tin	.0508	.080	.089
16	19 X 29	Copper	Tin	.057	.087	.095
16	26 X 30	Copper	Tin	.060	.090	.098
14	1 X 14	Copper	Tin	.0641	.094	.103
14	19 X 27	Copper	Tin	.072	.102	.110
14	41 X 30	Copper	Tin	.076	.084	.092
12	19 X 25	Copper	Tin	.091	.121	.129
12	37 X 28	Copper	Tin	.089	.119	.127
12	65 X 30	Copper	Tin	.093	.123	.131

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 5.0 kilovolts (kV).
Impulse dielectric test: 10.0 kV.
Tank test: 3.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C
K = 10,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 55°C ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
26 through 16	2
14, 12	3

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Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 150°C.
Flammability: Applicable - 30 seconds (maximum) burning time
3 inches (maximum) flame travel
Soldering: Applicable.
Heating aging: 25 percent change (maximum) 96 hours at 135°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):

M16878/	15	B	C	B	905
Military designation	Specification sheet	Conductor material	Conductor size	Conductor stranding	Color code

Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy -SH
(Project 6145-N300-14)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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